

ARUSH KUMAR JAIN

International Institute of Information Technology, Bangalore

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EDUCATION

International Institute of Information Technology, Bangalore

July 2025 – Present

Bachelor of Technology in Computer Science and Engineering

PROJECTS

Sicari.works | Node.js, Express, MySQL, Nginx, Azure, Cloudflare | [Live](#)

April 2026 – May 2026

- Built a full-stack criminal job board focused on robust, production-grade security. Fixers post heists, Sicarios apply based on skill matching.
- Created a custom skill-matching engine using array intersection to compute a weighted fit score (0–100%) server-side, sorting the heist feed by relevance per user.
- Secured with 8 layers of protection: HttpOnly JWT cookies, bcrypt (12 rounds) with legacy password auto-upgrade, timing-safe request signing via crypto.timingSafeEqual, append-only audit logs, Zod schema validation, parameterized queries, Helmet header hardening, and CORS whitelisting.
- Self-hosted on Azure VM, configured Nginx as reverse proxy, PM2 for process management with auto-restart, UFW firewall rules, and Cloudflare for DDoS protection, SSL termination, and Layer 7 proxying; automated deployments via GitHub Actions CI/CD.

Omen - Offline Mesh Emergency Network | C, UDP Sockets, POSIX Threads, libsodium, Docker | [GitHub](#)

May 2026

- Developed an offline peer-to-peer UDP mesh network in C with automatic device discovery, providing zero-infrastructure communication for disaster zones.
- Implemented a custom Reliable Data Transfer (RDT 3.0) protocol with stop-and-wait ACKs to guarantee critical message delivery across unstable wireless links.
- Secured node communication using *libsodium* authenticated encryption (XSalsa20-Poly1305) with password-derived symmetric keys (BLAKE2b), enabling rapid deployment without pre-shared key infrastructure.
- Designed a multi-threaded backend with POSIX threads to handle concurrent messaging, featuring a heartbeat monitor that automatically purges disconnected devices.

TECHNICAL SKILLS

Languages: C, Python, JavaScript, C++

Backend: Node.js, Express.js, REST APIs, JWT, bcrypt, Zod, MySQL, mysql2

DevOps & Infrastructure: Git, Azure, Nginx, PM2, Docker, Cloudflare, GitHub Actions CI/CD, Cloudinary

Security: libsodium, timing-safe operations, defence-in-depth sanitization, idempotency, audit logging

Concepts: Data Structures, Algorithms, Computer Networks, Computer Architecture

RELEVANT COURSEWORK

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|------------------|------------------|-------------------------|------------------------------|
| • Algorithms | • Calculus | • Computer Architecture | • Data Structures |
| • Linear Algebra | • Digital Design | • Computer Networks | • Probability and Statistics |

ACHIEVEMENTS

- JEE Main 2025:** Secured All India Rank 6080 (99.61 percentile) among 1.5 million+ aspirants.
- Hackathon:** Won 1st place at Hacknite 2026, IIIT Bangalore.

LEADERSHIP & EXTRACURRICULAR

Confluence Core | Design & Technical

Feb 2026 – Present

- Contributing as a core team member to IIIT Bangalore's flagship speaker series, overseeing visual production and orchestrating sessions in collaboration with distinguished industry leaders and founders.

IIIT-B Entrepreneurship Cell (E-Cell) | Design Team

Sep 2025 – Present

- Responsible for end-to-end design of E-Cell initiatives, encompassing visual assets, event creatives, and brand identity.